

Nicholas A. Henry

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EDUCATION

Northeastern University Boston, MA
Bachelor of Science in Mechanical Engineering and Computer Science May 2029
Coursework: Diff Eq and Linear Algebra, Discrete Structures, Program Design and Implementation, Statics GPA: 3.79
Activities: Northeastern Electric Racing FSAE, Students for the Exploration and Discovery of Space, Deans List
Studied a Semester Abroad at Queen's University Belfast in Northern Ireland

SKILLS

Software Skills: Python, TensorFlow, React, Next.js, Docker, Dash, Plotly, C#, ROS 2, Java, Git, JavaScript, HTML/CSS, NumPy, Flask, Linux, REST APIs, Supabase, SQL, GitHub Actions, WordPress, Vercel

Engineering Skills: Project Management, Fusion360, SolidWorks, AutoCAD, MATLAB, Problem Solving, Cross-Functional Collaboration, 3D Printing, Laser Cutting, CNC Mill, MIG and TIG Welding

Certifications & Training: IBM Artificial Intelligence Fundamentals, Microsoft C# Certification, GitHub Foundations

EXPERIENCE

Northeastern Electric Racing - Python, Dash, Plotly, React, Docker, Git Boston, MA
Lead Software Engineer for Simulation Team Jan 2026 – Present

- Lead on the simulation team developing a multi-body vehicle simulation tool for NER engineers to test and examine how changes to geometry and other variables go on to affect the speed and times of the car on custom created tracks
- Utilize Python as the backend to do all the physics with Dash, Plotly, and React for the frontend of the site to easily create all graphs and metrics making the UI easy for non-technical people to use the site effectively
- Conduct meetings and delegate work to newer members while also researching how to advance our physics engine

Students for the Exploration and Discovery of Space - Python, ROS 2, Git, Linux, Docker Boston, MA
Software Engineer for Network Exploration Swarm Toolkit Team Jan 2026 – Present

- Operate in NEST, a project to create a swarm of C.O.B.R.A. (Crater Observing Bio-inspired Rolling Articulator) snake robots to explore and develop multi-agent collaboration in order to complete teamwork based exploration tasks
- Developed the camera sensor ecosystem creating a simultaneous mapping program for multi-agent systems

SEONeeded - Wordpress, SEOMoz, Semrush Miami, FL
Web Optimization Engineer Oct 2025 – Aug 2026

- Perform client-facing tasks like identifying keywords, website maintenance, and strategies to target in order to drive more traffic for certain services increasing click-on rate and sales for 15 clients across the Greater South Florida region
- Adapted to the new wave of generative engine optimization space by learning what makes LLM models reference client's websites and how certain worded prompts get tailored responses, recommendations, and links to sources

Search Engine Optimization Intern Jun 2025 – Sep 2025

- Drove a 54% increase in website traffic through targeted optimization by leveraging WordPress and keyword research

PROJECTS

SheepFinder - React, Next.js, Vercel, SQL, Git May 2026 - Present

- Developed a Next.js and React web app, which I deployed through Vercel, based on my experience studying abroad in Northern Ireland designed to assist farmers and create an ecosystem where townspeople can report sightings of lost animals to make it easier for farmers to pick them up once escaped
- Spoke with farmers from Northern Ireland to gain an insight into what is the main issue and the best ways to solve them and built an app around that feedback like how sheep markings are designated across farms in different towns
- Have at least 20+ unique documented uses in Northern Ireland and heard from a farm of the benefits of the site

Portfolio Website - HTML/CSS, JavaScript, SQL, Supabase, GitHub Actions, Git Mar 2025 - Present

- Built a site with HTML/CSS and Javascript hosted on Github Pages with a custom domain to display myself and projects
- Applied SQL databases with Supabase in order to have a global comment section where visitors can write something and use GitHub Actions in order to autoupdate my resume on the site to match this document as it changes in Google Docs

Text Transcriber - Python, TensorFlow, Google APIs, Git Jul 2025

- Designed and trained an AI model using TensorFlow on 100,000+ character images to recognize and convert handwriting into text and to then convert into audio with Google TTS as part of a 3-person team
- Tested with a group of visibly impaired people to see and hear writing and signage enabling them to be less dependant on caretakers in everyday life, with successful results in the limited cohort of 88% understanding the responses